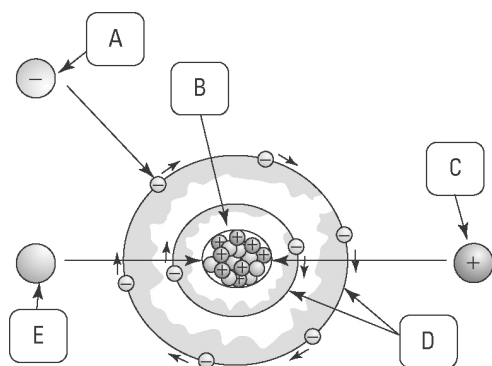


The structure of the atom

Student: Class:

The atom

1. The following drawing shows a simplified structure of an atom.

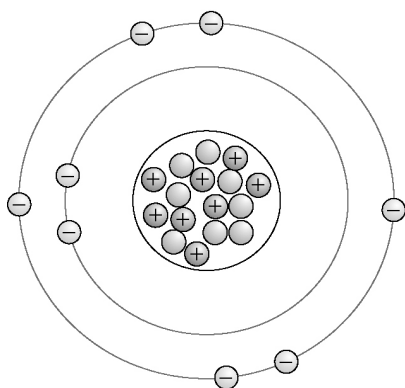


Identify the labelled components of the atom.

- A:
 B:
 C:
 D:
 E:

The oxygen atom

2. The diagram below shows the atomic structure of oxygen.



- (a) How many of each type of sub-atomic particle are present?

Electrons:

Protons:

Neutrons:

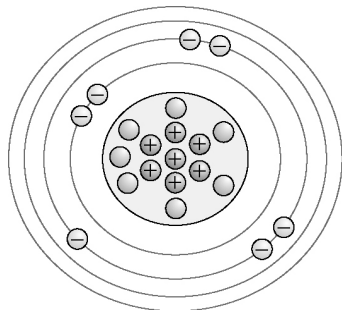
- (b) Is the oxygen atom shown above neutral or charged? Explain.

.....

- (c) How many nucleons (total particles) are present in the nucleus?

Atoms and ions

3. The diagram below shows a nitrogen atom. A nitrogen atom can form an ion by gaining three more electrons, so that its electron configuration becomes 2, 8.



- (a) In the diagram above, how many of each type of sub-atomic particle are present?

Electrons:

Protons:

Neutrons:

- (b) Draw the nitrogen ion described above.

- (c) In the diagram you have just drawn, how many of each type of sub-atomic particle are present?

Electrons:

Protons:

Neutrons: